



AVIGNON
UNIVERSITÉ

Exercices – Chapitre 3

Optimisation combinatoire quantique

Patrice SEBASTIANO

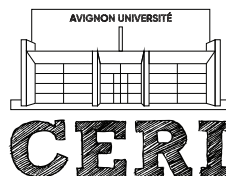
24/06/2026

**Master d'Informatique
Informatique Quantique**

UE Stage de Recherche – LIA Avignon

Encadrant
Serigne GUEYE

UFR
SCIENCES
TECHNOLOGIES
SANTÉ



CENTRE
D'ENSEIGNEMENT
ET DE RECHERCHE
EN INFORMATIQUE
ceri.univ-avignon.fr

Sommaire

Titre	1
Sommaire	2
1 Exercices du Chapitre 3	3

1 Exercices du Chapitre 3

Exercice 1 – p. 86

The maximum cut in a graph doesn't need to be unique. Find a maximum cut for the graph in *Figure 3.1* in which vertices 0 and 1 are in the same set.

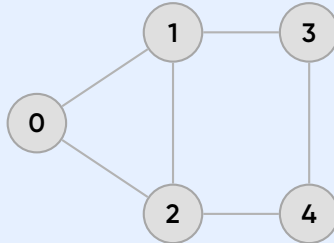
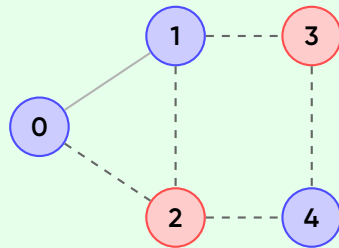


Figure 3.1 – Graph with 5 vertices and 6 edges

Solution

We assign $S_1 = \{0, 1, 4\}$ (blue) and $S_2 = \{2, 3\}$ (red). The only non-cut edge is $(0,1)$; all other 5 edges cross the partition, giving a **maximum cut of size 5**.



● $S_1 = \{0, 1, 4\}$

● $S_2 = \{2, 3\}$

--- cut edge (5 total)

— non-cut edge

Exercice 2 – p. 87

Write the Max-Cut problem for the graph in *Figure 3.3* (six vertices 0–5, edges $(0,1)$, $(0,2)$, $(1,2)$, $(1,4)$, $(2,3)$, $(3,4)$, $(3,5)$, $(4,5)$) as an optimization problem :

$$\text{Minimize } \sum_{(j,k) \in E} z_j z_k, \quad \text{subject to } z_j \in \{-1, 1\}.$$

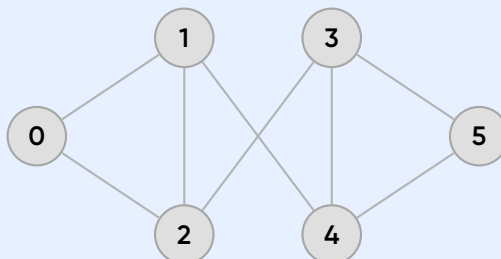


Figure 3.3 – Graph with 6 vertices and 8 edges

What is the value of the function to be minimized when $z_0 = z_1 = z_2 = 1$ and $z_3 = z_4 = z_5 = -1$? Is it an optimal cut?

Solution

Formulation. With $E = \{(0,1),(0,2),(1,2),(1,4),(2,3),(3,4),(3,5),(4,5)\}$ the objective is :

$$\text{Minimize } z_0z_1 + z_0z_2 + z_1z_2 + z_1z_4 + z_2z_3 + z_3z_4 + z_3z_5 + z_4z_5.$$

Evaluation at $z_0 = z_1 = z_2 = 1, z_3 = z_4 = z_5 = -1$.

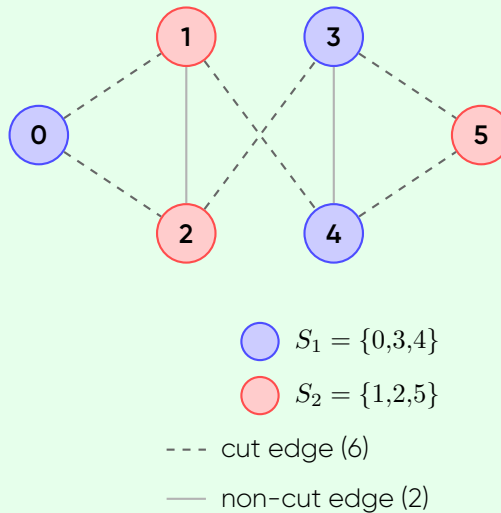
$$\underbrace{z_0z_1}_{+1} + \underbrace{z_0z_2}_{+1} + \underbrace{z_1z_2}_{+1} + \underbrace{z_1z_4}_{-1} + \underbrace{z_2z_3}_{-1} + \underbrace{z_3z_4}_{+1} + \underbrace{z_3z_5}_{+1} + \underbrace{z_4z_5}_{+1} = 4.$$

Only edges (1,4) and (2,3) are cut; the partition $\{0,1,2\}|\{3,4,5\}$ is not optimal.

Optimal cut. We assign $S_1 = \{0, 3, 4\}$ (blue) and $S_2 = \{1, 2, 5\}$ (red), i.e. $z_0 = z_3 = z_4 = +1$ and $z_1 = z_2 = z_5 = -1$. Edges (1,2) and (3,4) stay within a set ($z_jz_k = +1$); the other 6 cross the partition ($z_jz_k = -1$) :

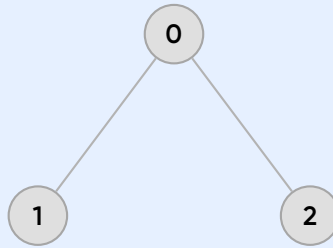
$$\text{objective} = 2 \times (+1) + 6 \times (-1) = -4.$$

Since $-4 < 4$, this strictly improves on the proposed assignment, which is therefore **not optimal**.



Exercise 3 – p. 93

Consider the graph below (three vertices, edges (0,1) and (0,2), no edge (1,2)).



Compute

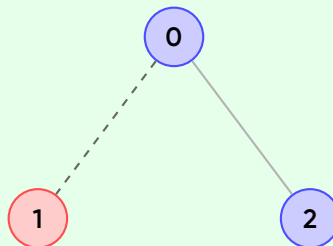
$$\langle 010 | (Z_0 Z_1 + Z_0 Z_2) | 010 \rangle \quad \text{and} \quad \langle 100 | (Z_0 Z_1 + Z_0 Z_2) | 100 \rangle.$$

Does any of those states minimize $\langle x | (Z_0 Z_1 + Z_0 Z_2) | x \rangle$?

Solution

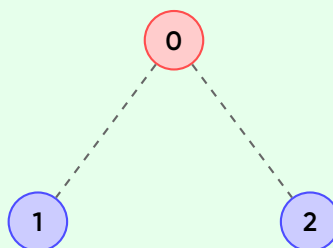
For a computational basis state, Z acts as $+1$ on $|0\rangle$ and -1 on $|1\rangle$, so the expectation value reduces to a product of eigenvalues along each edge.

State $|010\rangle$ – only edge $(0,1)$ is cut (-1), edge $(0,2)$ is not ($+1$):



$$\langle 010 | (Z_0 Z_1 + Z_0 Z_2) | 010 \rangle = -1 + 1 = 0.$$

State $|100\rangle$ – both edges $(0,1)$ and $(0,2)$ are cut (-1 each):



$$\langle 100 | (Z_0 Z_1 + Z_0 Z_2) | 100 \rangle = -1 + (-1) = -2.$$

Is $|100\rangle$ optimal? The graph has only 2 edges; each contributes at least -1 to the sum, so the minimum possible value is -2 . Since $|100\rangle$ achieves -2 , it **minimizes** the expectation value and is therefore optimal. $|010\rangle$ with value 0 is not.

Exercise 4 – p. 104

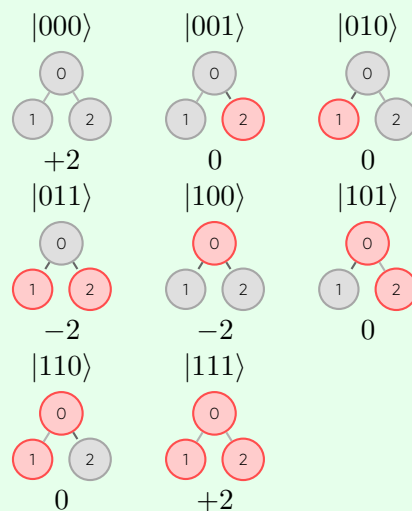
Write code to compute the expectation value of all the possible cuts of the graph in Figure 3.5 (three vertices 0, 1, 2 with edges $(0,1)$ and $(0,2)$). How many optimal solutions are there?

Solution

The code iterates over all $2^3 = 8$ computational basis states, builds each statevector via tensor product, and computes $\langle x | H_{\text{cut}} | x \rangle$ with `psi.expectation_value(H_cut)`.

Python – expectation values (Chapter 3, p. 104)

```
1 import itertools
2 from qiskit.quantum_info import Statevector, SparsePauliOp
3
4 zero = Statevector([1, 0])
5 one = Statevector([0, 1])
6 H_cut = SparsePauliOp("ZZI") + SparsePauliOp("ZIZ")
7
8 expectation_values = {}
9 for bits in itertools.product([0, 1], repeat=3):
10     psi = zero if bits[0] == 0 else one
11     for bit in bits[1:]:
12         psi = psi ^ (zero if bit == 0 else one)
13     expectation_values[bits] = psi.expectation_value(H_cut)
14
15 for bits, val in expectation_values.items():
16     print(f"|{''.join(map(str, bits))}> : {val.real:6.2f}")
```



The minimum value is -2 , achieved by **2 optimal solutions** : $|011\rangle$ and $|100\rangle$ (both cut all edges). This confirms the result from Exercise 3.3.

Exercise 5 – p. 109

Write the Subset Sum problem for $S = \{-1, -2, 3, -4\}$ and $T = 0$ as a QUBO problem, and transform it into an instance of the Ising model.

Solution

QUBO formulation. Assign a binary variable $x_j \in \{0, 1\}$ to each element $s_j \in S$. The

subset sum constraint $\sum_j s_j x_j = T = 0$ is enforced by minimizing its square :

$$\text{Minimize } f(x) = (x_0 - 2x_1 + 3x_2 - 4x_3)^2, \quad x_j \in \{0,1\}.$$

Expanding and using $x_j^2 = x_j$:

$$f(x) = x_0 + 4x_1 + 9x_2 + 16x_3 - 4x_0x_1 + 6x_0x_2 - 8x_0x_3 - 12x_1x_2 + 16x_1x_3 - 24x_2x_3.$$

Ising conversion. Substitute $x_j = \frac{1-z_j}{2}$ with $z_j \in \{-1, +1\}$ directly into the quadratic form :

$$f(x) = (x_0 - 2x_1 + 3x_2 - 4x_3)^2$$

Grouping the terms under a common denominator inside the square gives :

$$H = \left(\frac{(1 - z_0) - 2(1 - z_1) + 3(1 - z_2) - 4(1 - z_3)}{2} \right)^2$$

Expanding and simplifying the numerator yields :

$$H = \frac{1}{4} \left[-z_0 + 2z_1 - 3z_2 + 4z_3 - 2 \right]^2$$

Since $z_j^2 = 1$, all squared terms evaluate to constants ($1 + 4 + 9 + 16 + 4 = 34$) and can be dropped as they do not affect the minimizer.

Expanding the remaining cross-products (using the identity $(A + B + C + D + E)^2 = \sum \text{squares} + 2 \sum \text{cross-products}$) gives the interaction terms and the linear longitudinal fields.

Pair (i,j)	Cross-product term	Ising coupling (J_{ij})
(0,1)	$2(-z_0)(2z_1) = -4z_0z_1$	$-z_0z_1$
(0,2)	$2(-z_0)(-3z_2) = 6z_0z_2$	$\frac{3}{2}z_0z_2$
(0,3)	$2(-z_0)(4z_3) = -8z_0z_3$	$-2z_0z_3$
(1,2)	$2(2z_1)(-3z_2) = -12z_1z_2$	$-3z_1z_2$
(1,3)	$2(2z_1)(4z_3) = 16z_1z_3$	$4z_1z_3$
(2,3)	$2(-3z_2)(4z_3) = -24z_2z_3$	$-6z_2z_3$

The linear z_j coefficients are obtained from the cross-products between each variable and the constant term -2 , scaled by the global factor $\frac{1}{4}$:

$$\begin{aligned} z_0 : \frac{2(-z_0)(-2)}{4} &= +z_0, & z_1 : \frac{2(2z_1)(-2)}{4} &= -2z_1, \\ z_2 : \frac{2(-3z_2)(-2)}{4} &= +3z_2, & z_3 : \frac{2(4z_3)(-2)}{4} &= -4z_3. \end{aligned}$$

The final Ising Hamiltonian (up to a global constant) is :

$$\begin{aligned} H = & -z_0z_1 + \frac{3}{2}z_0z_2 - 2z_0z_3 - 3z_1z_2 + 4z_1z_3 - 6z_2z_3 \\ & + z_0 - 2z_1 + 3z_2 - 4z_3. \end{aligned}$$

Exercise 6 – p. 115

Consider objects with values $v = (3, 1, 7, 7)$ and weights $w = (2, 1, 5, 4)$. Write the Knapsack problem for the case in which the maximum weight is $W = 8$ as a binary linear program.

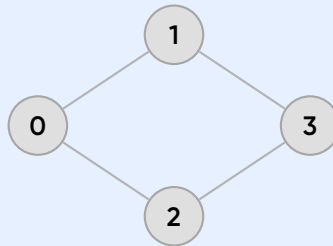
Solution

With $v = (3, 1, 7, 7)$, $w = (2, 1, 5, 4)$, $W = 8$, and binary variables $x_j \in \{0, 1\}$:

Minimise $-3x_0 - x_1 - 7x_2 - 7x_3$, subject to $2x_0 + x_1 + 5x_2 + 4x_3 \leq 8$.

Exercise 7 – p. 119

Consider the graph below (four vertices, edges $(0,1)$, $(0,2)$, $(1,3)$, $(2,3)$).



Write the QUBO version of the problem of checking whether the graph is 2-colorable.

Solution

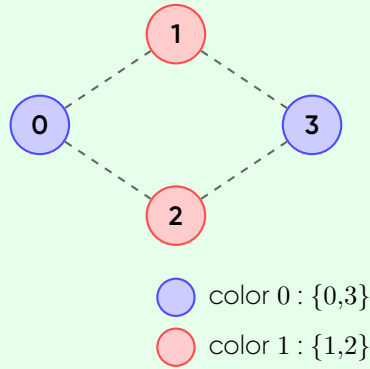
Introduce binary variables $x_{jl} \in \{0, 1\}$ where $x_{jl} = 1$ means vertex j receives color $l \in \{0, 1\}$. Two penalty terms enforce the constraints :

- **Each vertex gets exactly one color** : $\sum_{l=0}^1 x_{jl} = 1$, penalised as $\left(\sum_{l=0}^1 x_{jl} - 1\right)^2$.
- **Adjacent vertices get different colors** : for each edge (j, h) and each color l , x_{jl} and x_{hl} cannot both be 1, penalised as $x_{jl}x_{hl}$.

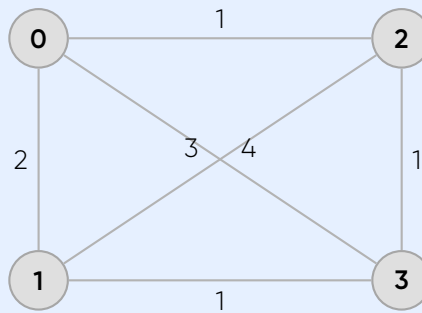
The full QUBO is :

$$\text{Minimise } \sum_{j=0}^3 \sum_{l=0}^1 (x_{jl} - 1)^2 + \sum_{(j,h) \in E} \sum_{l=0}^1 x_{jl} x_{hl}, \quad x_{jl} \in \{0, 1\}.$$

The graph is 2-colorable if and only if the minimum is 0. Here the optimum is achieved by $x_{0,0} = x_{3,0} = x_{1,1} = x_{2,1} = 1$ (all other $x_{jl} = 0$), confirming the graph is 2-colorable (the 4-cycle is bipartite).


Exercise 8 – p. 121

Consider the following weighted graph (Figure 3.7) with 4 cities and symmetric edge costs :



Obtain the QUBO expression for the route cost in the TSP problem. The cost is symmetric ($w_{jk} = w_{kj}$) and the formulation uses binary variables $x_{jl} \in \{0,1\}$ ($x_{jl} = 1$ means city j is visited at step l), with $m = 3$ steps and cities $j \in \{0,1,2,3\}$:

$$\text{Minimise } \sum_{l=0}^{m-1} \sum_{j=0}^m \sum_{k=0}^m w_{jk} x_{jl} x_{k,l+1} + B \left(\sum_{l=0}^m x_{jl} - 1 \right)^2 + B \left(\sum_{j=0}^m x_{jl} - 1 \right)^2, \quad x_{jl} \in \{0,1\}.$$

Solution

The edge set with weights is $E = \{(0,2,1), (0,1,2), (2,3,1), (1,3,1), (0,3,3), (2,1,4)\}$ (notation : (j,k,w_{jk})). The symmetric weight matrix is :

$$W = \begin{pmatrix} 0 & 2 & 1 & 3 \\ 2 & 0 & 4 & 1 \\ 1 & 4 & 0 & 1 \\ 3 & 1 & 1 & 0 \end{pmatrix}.$$

The route-cost term (first sum) counts the cost of travelling from the city visited at step l to the city visited at step $l+1$. Expanding over all (j,k) pairs with $w_{jk} > 0$ and

steps $l \in \{0,1,2\}$ (indices mod m for the return leg) :

$$\begin{aligned} C_{\text{route}} = & 2(x_{0,0}x_{1,1} + x_{0,1}x_{1,2} + x_{0,2}x_{1,0}) \\ & + (x_{0,0}x_{2,1} + x_{0,1}x_{2,2} + x_{0,2}x_{2,0}) \\ & + 3(x_{0,0}x_{3,1} + x_{0,1}x_{3,2} + x_{0,2}x_{3,0}) \\ & + 4(x_{2,0}x_{1,1} + x_{2,1}x_{1,2} + x_{2,2}x_{1,0}) \\ & + (x_{2,0}x_{3,1} + x_{2,1}x_{3,2} + x_{2,2}x_{3,0}) \\ & + (x_{1,0}x_{3,1} + x_{1,1}x_{3,2} + x_{1,2}x_{3,0}), \end{aligned}$$

where symmetry ($w_{jk} = w_{kj}$) means each undirected edge contributes in both directions and the factor of 2 is already absorbed into the coefficients above.